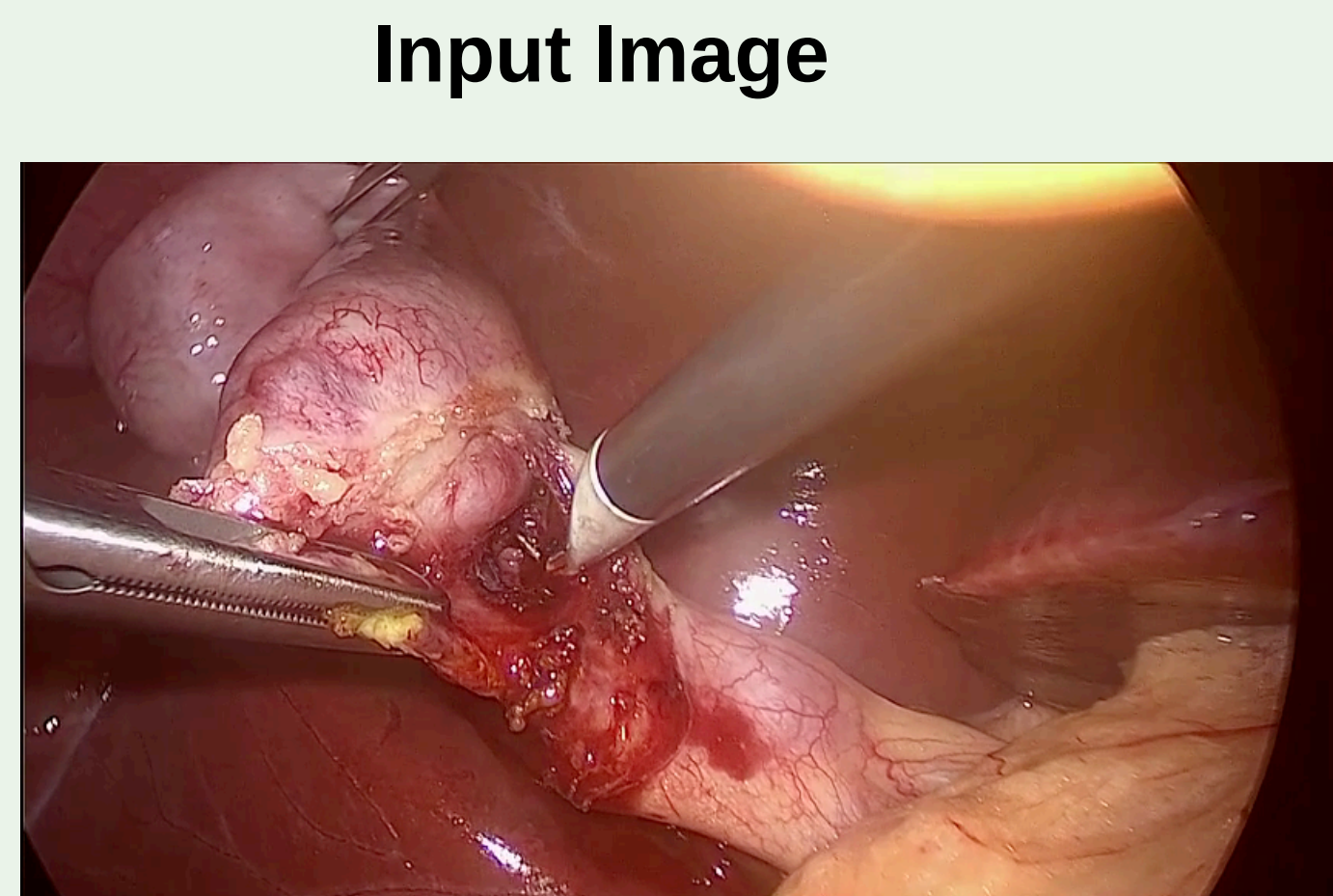
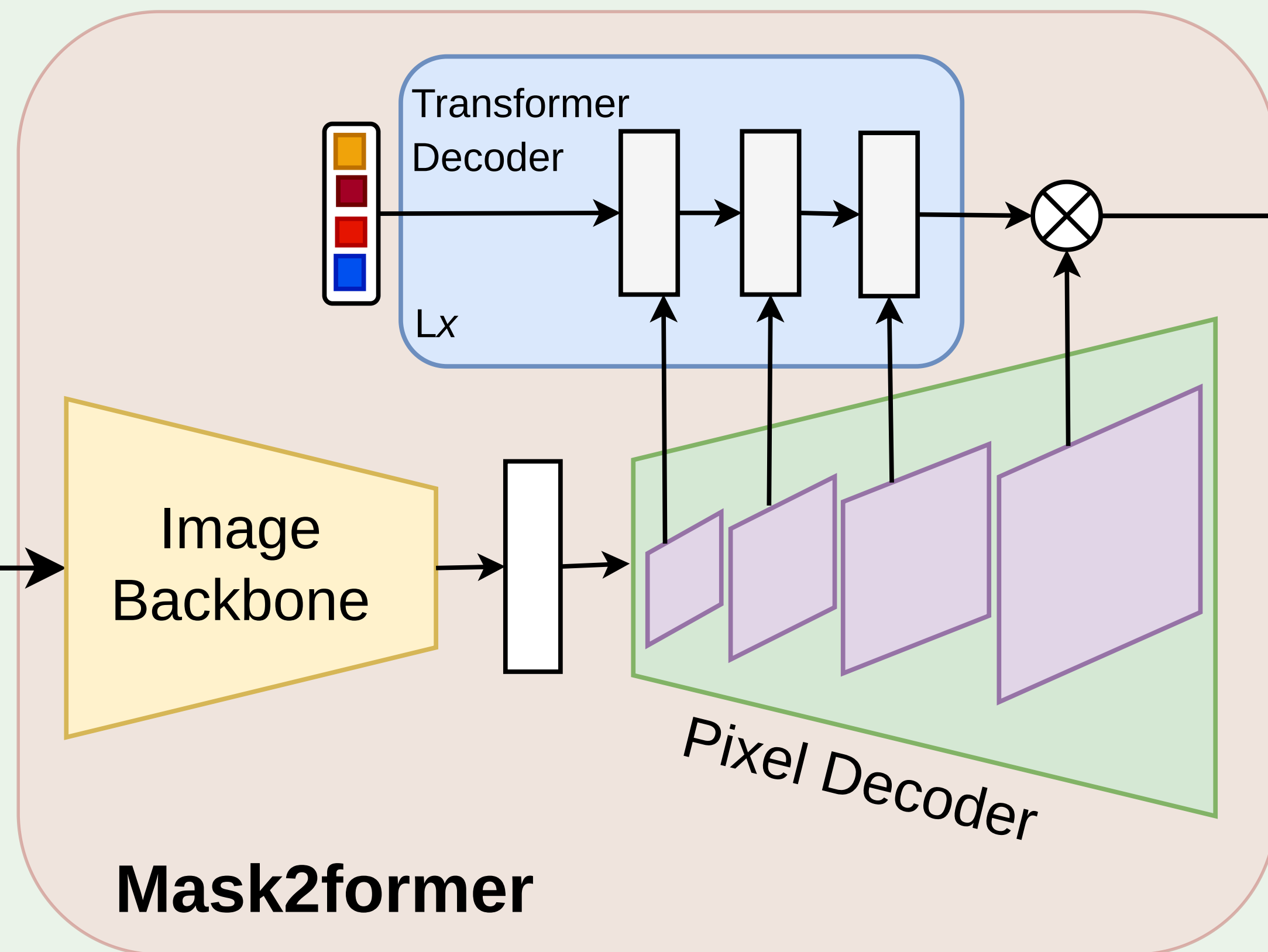


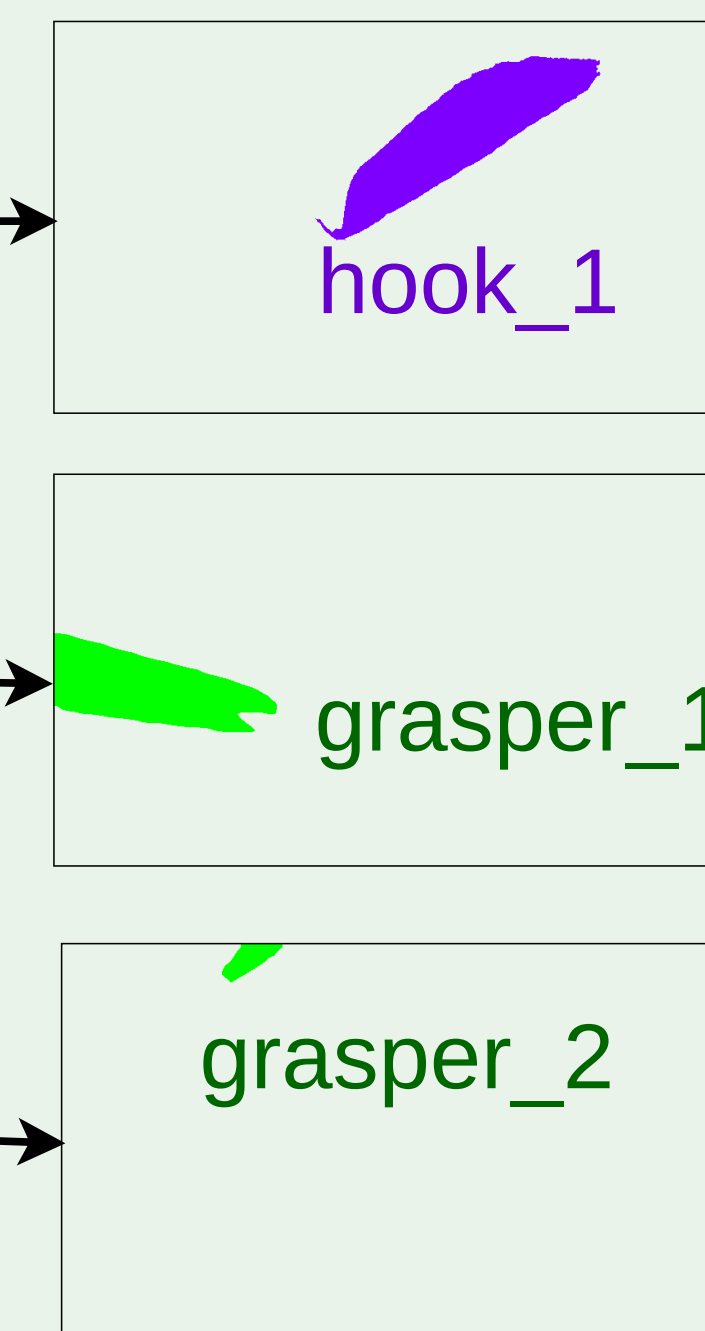


Input Image

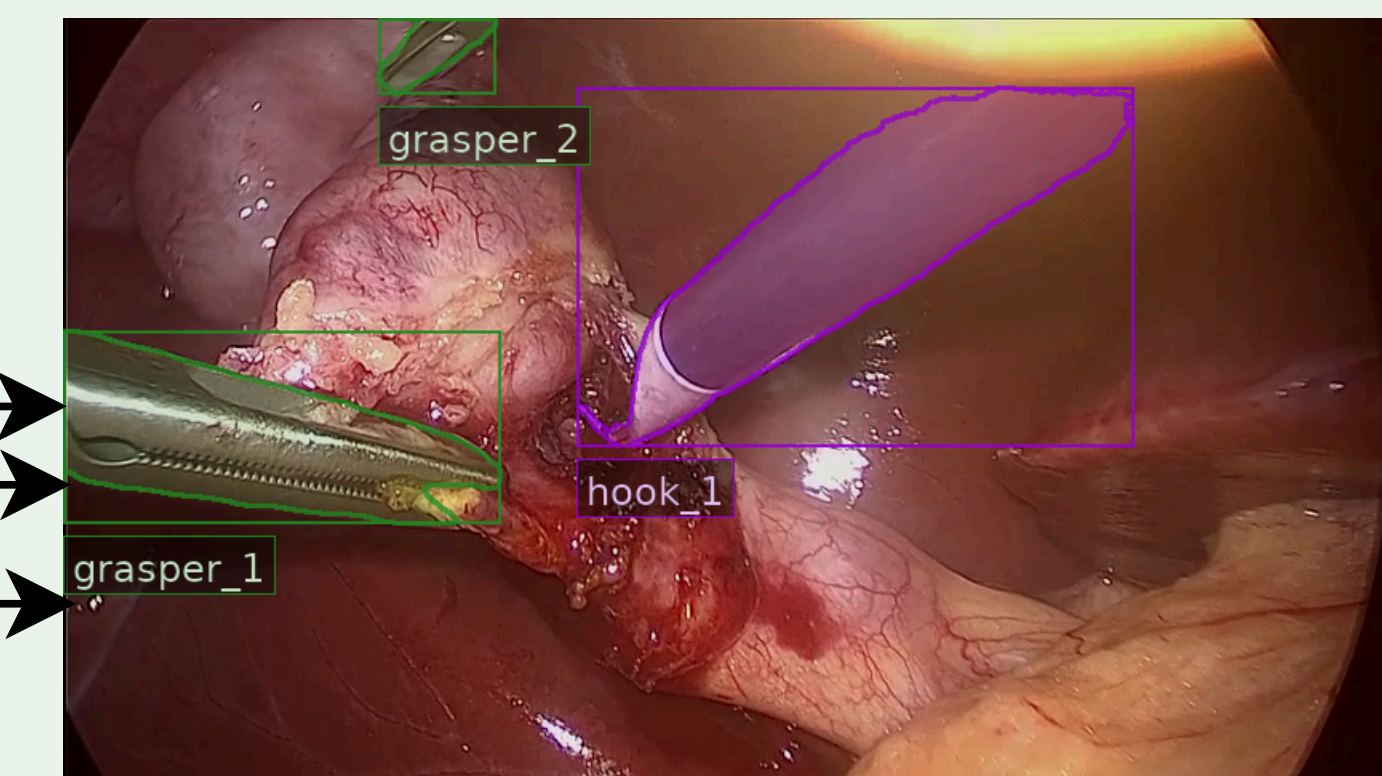


Mask2former

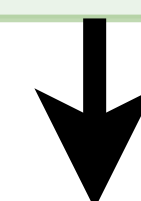
Instrument Instance Predictions



Instance Overlaid Surgical Image

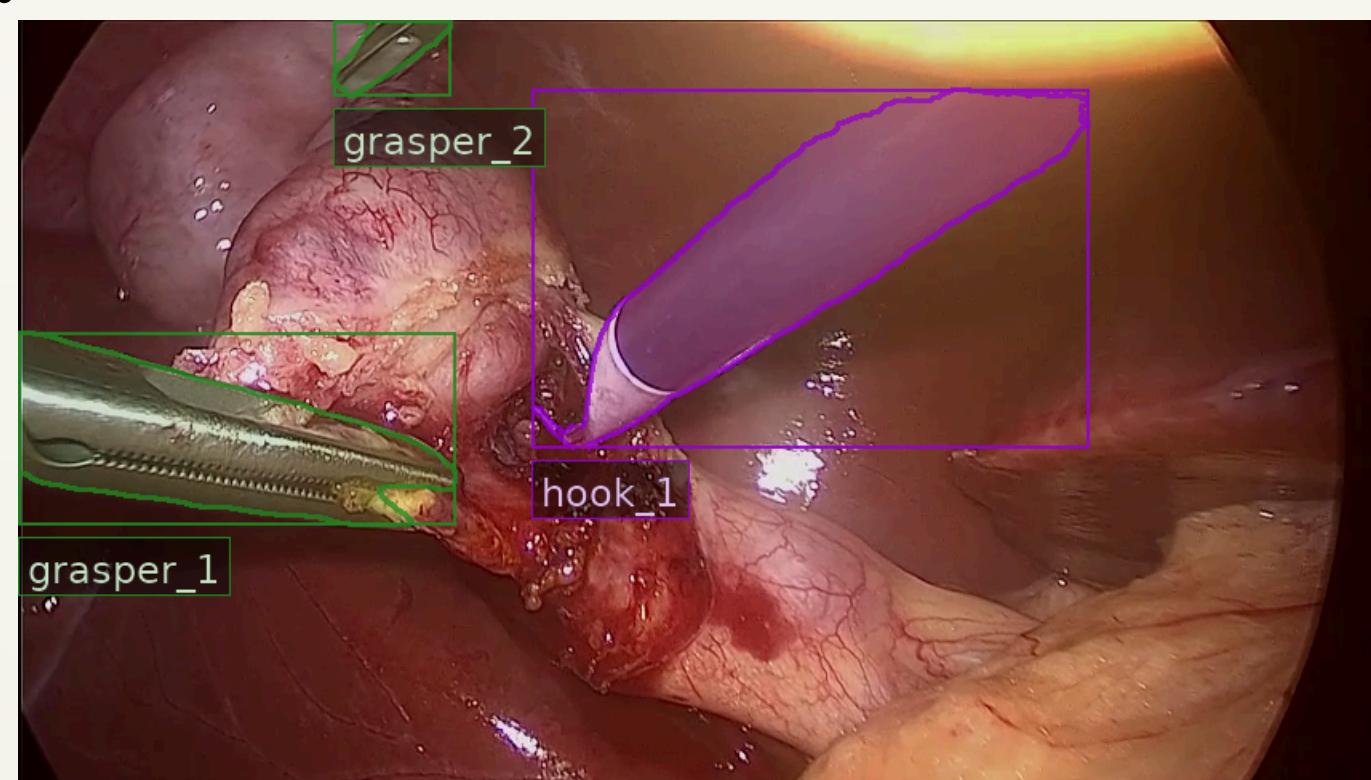


Stage 1: Instrument Grounding



Inputs to Qwen3-VL

* Instance Overlaid Surgical Image



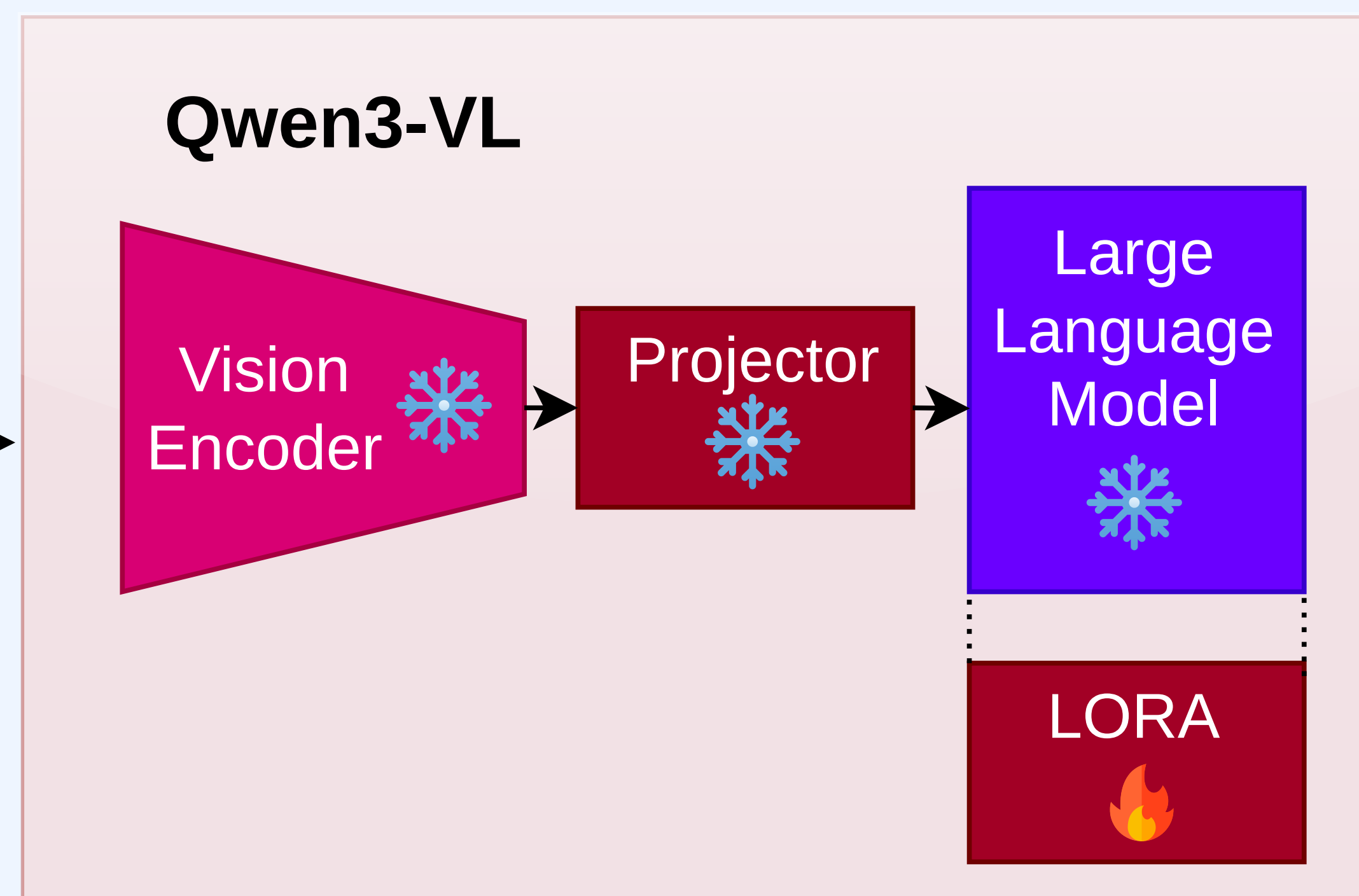
Prompt

For each instrument instance, assign exactly one (instrument, action, target) triplet using visual evidence.

...
Required output format (example):

```
{{
  "predictions": [
    {{ "id": "grasper_1", "instrument": "Grasping Forceps", "action":
"retract", "target": "gallbladder" }},
  ]
}}
```

Qwen3-VL



Predicted Grounded Surgical Triplets

Mask	Instrument Instance	Verb	Target
	hook_1	dissect	cystic_duct
	grasper_1	retract	cystic_duct
	grasper_2	retract	gallbladder

interactions

Stage 2: Multimodal Interaction Reasoning